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Modification of empirical core loss calculation methods including flux distribution

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Abstract: Using medium-frequency high-power transformers leads to significant decrease in size and volume of high-power converters in different applications such as offshore wind farms and electric traction. However, increasing the operating frequency leads to several challenges such as increase in the core loss. Consequently, estimating magnetic loss with high accuracy is essential especially in thermal modelling point of view and temperature rise calculation. The main contribution of this study is to introduce a novel and accurate model to investigate the spatial distribution of flux density inside the core. The distribution profile depends on the characteristics of the magnetic material and the configuration of the core. The accuracy of this method has been compared with the experimental results of other studies. Variation of flux density and core loss also depend on the operating magnetisation where at higher inductions, the difference between the minimum and maximum will be decreased. Owing to inhomogeneous distribution of flux, higher portion of the core loss will be generated in the inner half of the core. Ultimately, it has been expressed that considering the non-even distribution of the flux results in higher maximum temperature of the core.

1 Introduction

Magnetic components are the bulkiest part of high-power converters. In recent decades, medium-frequency transformer solution seems to be beneficial since the volume of the magnetic part is inversely proportional to the frequency of the exciting voltage [1]. The range of fundamental frequency of these transformers is typically from 500 Hz to 20 kHz [2]. This approach can thus save weight and cost of the converters. Besides, decreasing the weight of the converters results in reducing the complexity and cost of the wind turbine foundation and tower. Nevertheless, because of the increased operating frequency and also non-sinusoidal waveforms and harmonics, the losses inside the transformer can be increased drastically [3, 4]. Magnetic loss is one of the main categories of these losses which is generated in the core material. Owing to complexities of microscopic methods, there are several macroscopic approaches dealing with core loss calculation. These methods can be divided into two main groups, loss separation and empirical methods [5]. The former methods are insufficient and impractical for the designers because of complexity and the necessity of large amount of data resources of materials. Consequently, interests towards empirical methods has been increased which are easy to use and require few amount of measurements. All the above mentioned methods are based on total flux density induced in the magnetic core which is calculated by the integration of exciting voltage. Owing to the non-linear behaviour of magnetic materials and difference in magnetic path, the flux will not distribute homogonously in the core. This inhomogeneity should be considered in core loss estimation. The difference between the maximum and minimum of distributed flux depends on the behaviour of the material, configuration and dimensions of the core. Several experimental works have been done on toroidal [6], single-phase core type [7] and three-phase five leg transformers [8] dealing with the study of flux distribution in the core. An analytical solution to use a generic two-dimensional (2D) finite-difference method to accurately calculate the quasi 3D spatial distribution of loss in stacked power and distribution transformer core is described in [9]. Grimmond *et al.* [10] introduced an analytical method to study the magnetic properties of wound toroidal cores and flux distribution. The main drawback of their proposed method is that the magnetisation current is the input and flux intensity will be obtained by this current. However in reality, magnetising current is an unknown parameter and the exciting voltage is the known input. Accordingly, there is a lack of accurate analytical methods for core loss calculation which consider the variation of flux density in the core using exciting voltage.

In this paper, a comprehensive and easy to use method is explained in detail to evaluate the distribution of flux in the core and incorporate the changes of magnetic flux density in core loss calculation. Therefore in Section 2, a brief summary of different macroscopic core loss calculation methods is presented. Subsequently, the proposed method to determine the variation of flux in the core will be introduced and the results are compared with the several measurements implemented by other papers. In addition, this method is applied to different core topologies for the

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core loss calculation methods which were introduced in the Section 2. Finally, to investigate the importance of considering flux variation, the obtained method were implemented and applied in thermal modelling expressions to find the maximum temperature in the core.

2 Core loss calculation methods

The existing macroscopic core loss calculation methods can be categorised into two main groups, loss separation and empirical method. In the first method, the magnetic loss will be subdivided into three main components, hysteresis loss, classical eddy current loss and excessive loss [11]. This method has high capability in calculating core loss for non-sinusoidal excitations and harmonic environment

$$P = P_{\mathrm{h}} + \frac{1}{T}\frac{1}{\rho}\frac{\sigma d^2}{12}\left(\frac{\mathrm{d}B}{\mathrm{d}t}\right)^2 \mathrm{d}t + \frac{1}{T}K_{\mathrm{exc}}\left(\left|\frac{\mathrm{d}B}{\mathrm{d}t}\right|\right)^2 \mathrm{d}t \quad (1)$$

where P_{h} is the hysteresis loss, T is the period of flux density, σ and d are the conductivity and thickness of strips or laminations and ρ is the density of the magnetic material. For sinusoidal excitation, it will be [11]

$$P = K_{\mathrm{h}}\, B_m^{a\cdot B_{\mathrm{m}}^2 + b\cdot B_{\mathrm{m}} + c} f + K_{\mathrm{e}}\, B_{\mathrm{m}}^2 f^2 + 8.76\, K_{\mathrm{exc}}\, B_{\mathrm{m}}^{1.5} f^{1.5} \quad (2)$$

where f is the fundamental frequency, B_{m} is the maximum flux density. Nevertheless, obtaining the factors and coefficients (K_{h}, a, b, c, K_{e} and K_{exc}) in this method requires extensive measurements, which make it impractical for designers who prefer easy-to-use methods. The second group that is attractive for designers is empirical methods which initiated in 1892 by Steinmetz [12] introducing an equation to determine the magnetic loss, which was only dependent on the peak value of the magnetic induction. Afterwards, a more general equation was utilised, which also considers the frequency dependency of loss. It is known as original Steinmetz equation (OSE), which is essentially based on the curve fitting of several measurements under sinusoidal excitations at various frequencies and peak magnetic inductions [13]

$$P_{\mathrm{c}} = k f^{\alpha} B_{\mathrm{m}}^{\beta} \quad (3)$$

where k, α and β are unknown coefficients, which will be determined by manufacturer datasheets without requiring any detailed knowledge of the materials. The major restriction of this method is related to non-sinusoidal waveforms. In addition, it is not suitable for distorted B which happens in non-linear magnetic material and the coefficients are valid for a narrow range of frequency. To overcome this problem, some modifications have been introduced in recent years to increase the accuracy in harmonics environment. First, by establishing a relation between magnetic loss and the rate of change of the magnetic flux density, an equivalent frequency f_{eq} was introduced and used in modified Steinmetz equation (MSE) [14].

$$f_{\mathrm{eq}} = \frac{2}{(B_{\max} - B_{\min})^2 \pi^2}\int_0^{\mathrm{T}} \left(\frac{\mathrm{d}B(t)}{\mathrm{d}t}\right)^2 \mathrm{d}t \Rightarrow$$
$$P_c = \left(k\, f_{\mathrm{eq}}^{\alpha-1}\, B_{\mathrm{m}}^{\beta}\right) f_{\mathrm{r}} \quad (4)$$

where $B_{\max}$ and $B_{\min}$ are the maximum and minimum values of the flux density and f_{r} is the remagnetisation frequency. In [15], another idea – generalised Steinmetz equation (GSE) – was proposed supposing the fact that the core loss not only depends on the rate of the flux changing but also on the instantaneous value of the flux density

$$P_{\mathrm{c}} = \frac{1}{T}\int_0^{\mathrm{T}} k_i \left|\frac{\mathrm{d}B(t)}{\mathrm{d}t}\right|^{\alpha} |B(t)|^{\beta-\alpha}\, \mathrm{d}t \quad (5)$$

where k_i is

$$k_i = \frac{k}{(2\pi)^{\alpha-1}\int_0^{2\pi} |\cos(\theta)|^{\alpha} |\sin(\theta)|^{\beta-\alpha}\, \mathrm{d}\theta} \quad (6)$$

This method does not consider the history of the flux inside the material, which is critical in case of minor loops and reversal curves. Hence, another expression was introduced substituting the instantaneous value with the peak-to-peak value to be able to include the minor loops, known as improved generalised Steinmetz equation (IGSE) [16]

$$P_{\mathrm{c}} = \frac{1}{T}\int_0^{\mathrm{T}} k_{i_\mathrm{IGSE}} \left|\frac{\mathrm{d}B(t)}{\mathrm{d}t}\right|^{\alpha} (B_{\max} - B_{\min})^{\beta-\alpha}\, \mathrm{d}t \quad (7)$$

and k_{i_IGSE} is

$$k_{i_\mathrm{IGSE}} = \frac{k}{(2\pi)^{\alpha-1}\int_0^{2\pi} |\cos(\theta)|^{\alpha}\, 2^{\beta-\alpha}\, \mathrm{d}\theta} \quad (8)$$

In addition to the described verifications of Steinmetz equation, a correcting factor was introduced in [4] to match the OSE with non-sinusoidal waveforms. Also, equivalent elliptical loop model present a dynamic time-domain model combining the idea of both empirical and loss separation methods [17].

In all these described methods, the magnetic flux density inside the core is obtained by integration of the applied voltage $v(t)$.

$$B(t) = \frac{1}{NA_{\mathrm{c}}}\int_0^t v(t)\, \mathrm{d}t \quad (9)$$

where N is the number of winding turn and A_{c} is the cross section of the core. The resulted waveform of $B(t)$ in (9) is assumed as the average of the flux inside the core and is used for the estimation of the core loss. Consequently, the non-uniform distribution of the flux density is not considered. The main contribution of this paper is to find a simple and straightforward method to incorporate this flux variation into the core loss estimation procedure.

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3 Proposed methodology and verification

At medium-frequency high-power applications, new ferromagnetic materials such as nanocrystalline [18] and amorphous [19] have been used in the recent decades. These cores are made of very thin ribbons with the thickness of 20–25 μm to reduce the eddy current losses. Using these thin strips can result in lower eddy currents and better distribution of flux in the core. However, because of the difference of magnetic path lengths, the flux magnitude will be different in each ribbon. Two different designs of single-phase transformer cores are studied: *C*-core and toroidal core. The models of these cores are illustrated in Fig. 1. To find an expression of flux changes, the cross section of the core is descretised into several equal segments with the thickness of dx (dr in ring core) and the width of b. The summation of the fluxes in each path is equal to the total flux obtaining by the integration of applied voltage

$$\varphi_1 + \cdots + \varphi_i + \cdots + \varphi_n = \varphi_t = \frac{1}{N}\int v(t)\,\mathrm{d}t \overset{\sin}{\Rightarrow} \frac{V_{\mathrm{rms}}}{\sqrt{2}\,\pi N f} \quad (10)$$

where φ_i is the flux in ith path, φ_t is the total flux and V_{rms} is the root-mean-square value of exciting voltage. On the other hand, all the paths are subjected to the same ampere-turn.

$$R_1\varphi_1 = \cdots = R_i\varphi_i = \cdots = R_n\varphi_n = R_t\varphi_t = N \cdot I_{\mathrm{mag}} \quad (11)$$

where I_{mag} is the magnetising current, R_i is the reluctance of the ith path and R_t is the total reluctance of the core. By substituting (11) into (10), flux passing in the ith segment will be obtained

$$\varphi_i R_i \underbrace{\left(\frac{1}{R_1} + \cdots + \frac{1}{R_i} + \cdots + \frac{1}{R_n}\right)}_{1/R_t} = \varphi_t \;\Rightarrow\; \varphi_i = \frac{R_t\varphi_t}{R_i} \quad (12)$$

As can be seen, φ_i depends on total reluctance of the core and self-reluctance of the ith segment. R_i is represented as

$$R_i = \frac{l_i}{\mu_i A_i}, \quad \begin{cases} A_i = b \quad dx \;:\; \text{core type} \\ A_i = h_{\mathrm{r}} \quad dr \;:\; \text{ring core} \end{cases} \quad (13)$$

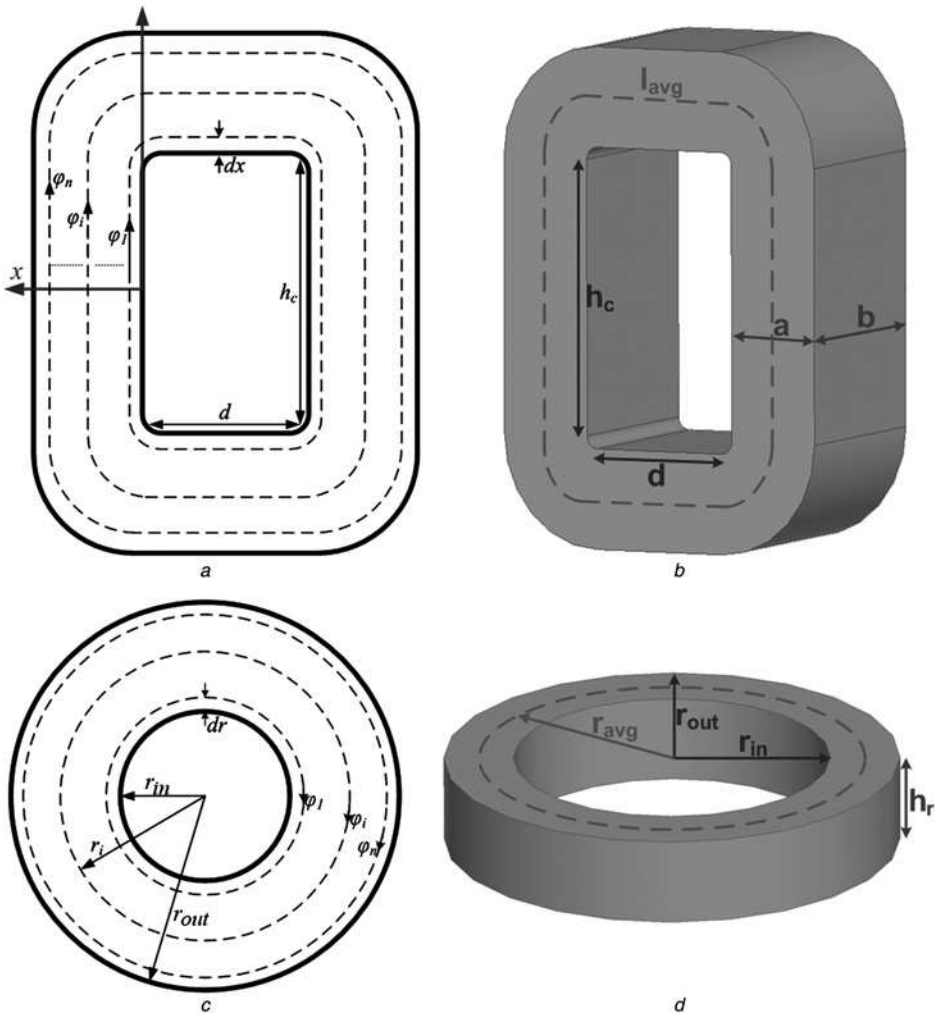


Fig. 1 *Two studied core configurations*

a Descritised model
b Geometrical dimensions of core type
c Descritised model
d Geometrical dimensions of ring core

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where A_i is the cross-section of each segment, b is the width of the core in core type, h_r is the height of ring core, μ_i is the permeability of the ith magnetic path and dx and dr are the control section of x- and r-axes.

Considering Figs. 1*a–d*, the magnetic length l_i will be specified as

$$l_i = \begin{cases} 2(d + h_c) + 2\pi\, i\, \mathrm{d}x & : \text{ core type} \\ 2\pi\,(r_{in} + i\,\mathrm{d}r) & : \text{ ring core} \end{cases} \quad (14)$$

where d, h_c and r_{in} are illustrated in Fig. 1. Hence, considering $r = i{\cdot}\mathrm{d}r$ and $x = i{\cdot}\mathrm{d}x$ and substituting (14) into (13), the localised value of magnetic flux density is

$$B(x) = \begin{cases} \dfrac{\mu_i\, \varphi_t\, R_t}{2(d + h_c) + 2\pi x} & : \text{ core type} \\ \dfrac{\mu_i\, \varphi_t\, R_t}{2\,\pi\,(r_{in} + r)} & : \text{ ring core} \end{cases} \quad (15)$$

where the total reluctance of the core is equal to

$$R_t = \frac{1}{\sum(1/R_i)} = \begin{cases} \dfrac{1}{\int_0^a \left((\mu_i\, b\, \mathrm{d}x)/(2(d + h_c) + 2\pi x)\right)} & : \text{ core type} \\ \dfrac{1}{\int_0^{r_{out} - r_{in}} \left((\mu_i\, h_r\, \mathrm{d}r)/(2\,\pi\,(r_{in} + r))\right)} & : \text{ ring core} \end{cases} \quad (16)$$

where a, h_r, r_{in} and r_{out} are shown in Fig. 1. As can be concluded from (15) and (16), the distribution of the flux density is dependent on the permeabilities of all segments. In Fig. 2, the measured relative permeability (μ_r) of nanocrystalline material is shown as a function of induction. With respect to this figure, μ_r varies strongly by changing the induced flux because of the non-linear characteristic and can have the value in the range of 28 000–1. Therefore there are two main factors that have substantial influence on

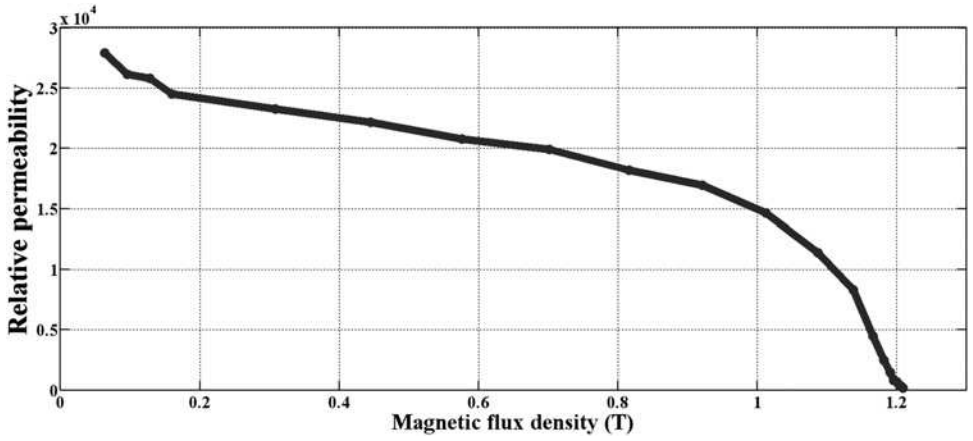


Fig. 2 *Dependency of measured relative permeability vs. induced flux density of VITROPERM 500 F at 5 kHz*

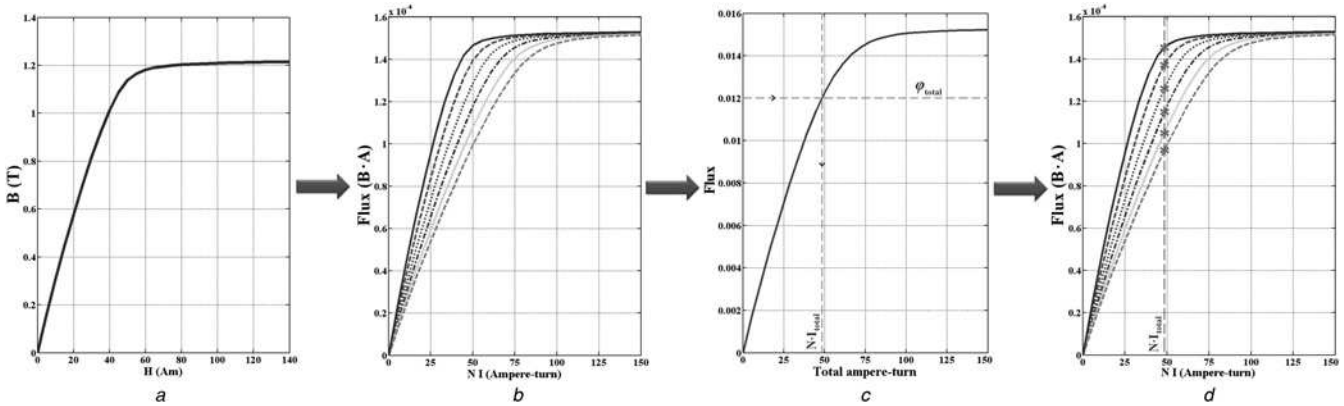


Fig. 3 *Four main steps of proposed method*

a Measured magnetisation curve of nanocrystalline
b φ-*NI* curve of each segment of core
c Total equivalent φ-*NI* curve of the core
d Induced flux in each segment by finding NI_{total} from *c*

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the flux displacement, difference in the magnetic length and the non-linear behaviour of the segments.

The proposed graphical method of this paper overcomes the complexity because of the non-linear effect. The methodology of the proposed method in this paper is presented in Fig. 3. First of all, a B-H curve of the magnetic material that describes the non-linear behaviour is required. The B-H curve in Fig. 3*a* is obtained by measurements on thin ring cores made of nanocrystalline material (VITROPERM 500 F). These measurements were done with electrical steel tester (Brockhaus MPG 100D), which is capable of testing ring cores at frequencies up to 10 kHz and polarisations up to 2 T. Afterwards, the core is divided into several subsections with similar thickness of d*x* (or d*r* in ring core) and width of *b*. The magnetic circuit equivalent of the winding with the separated segments of the core is depicted in Fig. 4 where R_i represents the reluctance of *i*th magnetic path as shown in Figs. 1*a* and *c*. These parallel paths have distinct characteristics because of different magnetic length. Therefore considering (17), by multiplying the flux density to the cross section and the magnetic length to the flux intensity (H), flux (φ) of each core segment as a function of ampere-turn (NI) will be obtained (Fig. 3*b*). All these segments are subjected to the same ampere-turn and according to the circuit shown in Fig. 4, all the segments are in parallel. Accordingly, the equivalent φ-NI curve can be achieved adding the flux values in each ampere-turn which results in Fig. 3*c*

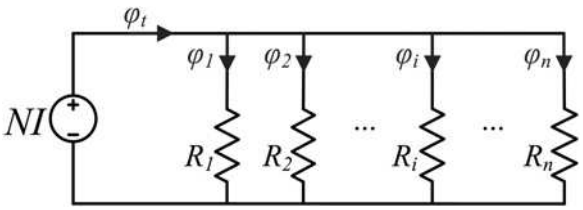


Fig. 4 *Equivalent magnetic circuit of the core considering all segments and induced ampere-turn*

$$\varphi_i = B_i A_i, \qquad H_i l_i = NI \tag{17}$$

The total flux determined by (10) gives the total ampere-turn (NI_{total}) using Fig. 3*c*. As was explained previously, all the core subsections are subjected to the same ampere-turn. Thus, as is illustrated in Fig. 3*d*, by finding the total ampere-turn, the induced flux and the flux density in each segment ($\varphi_1 - \varphi_n$ and $B_1 - B_n$) will be determined (see Fig. 3*d*).

To evaluate the proposed method, experimental results from two other papers are used. The first one is

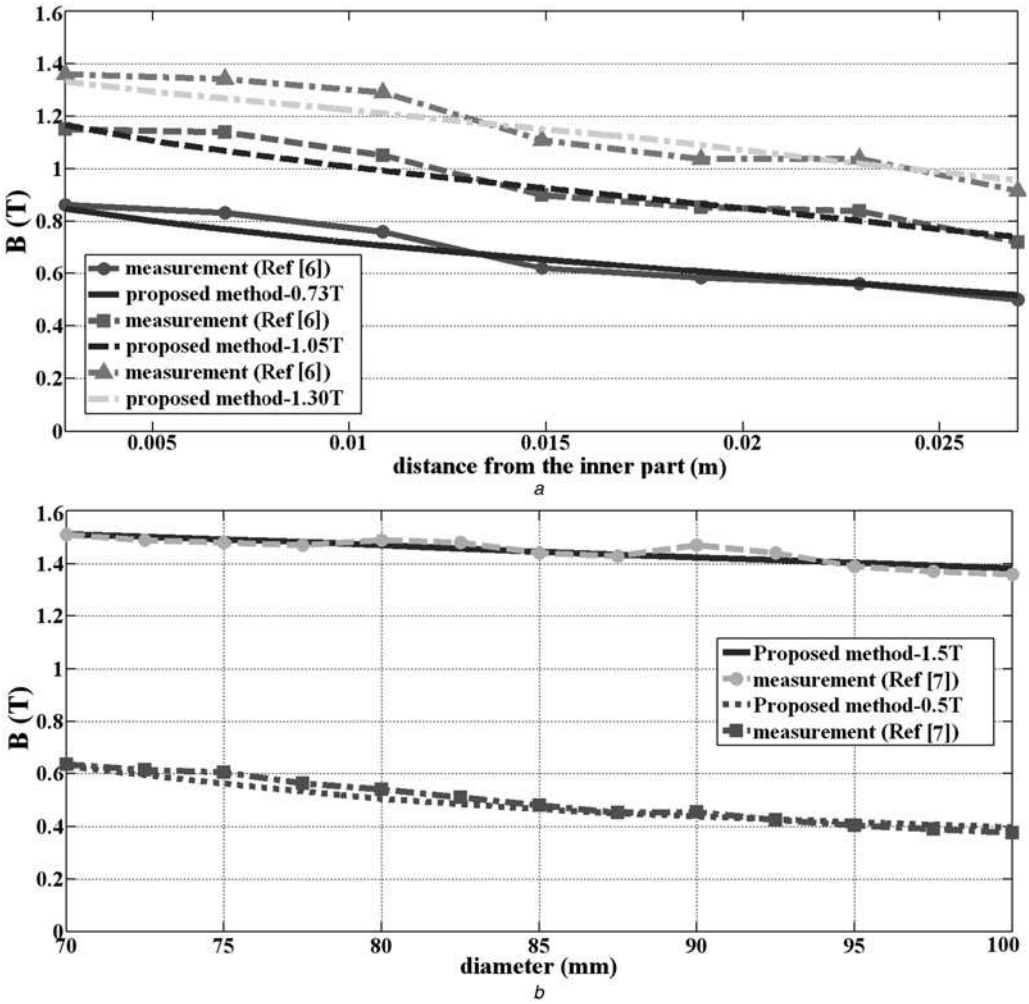


Fig. 5 *Comparison of experimental results of two references with proposed method*

a Core type [6]
b Ring core [7]

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implemented by Loizos *et al.* [6]. In this work, experiments were done on a single phase wound transformer core of high permeability Si–Fe steel with the height of 183 mm, thickness of 24.3 mm and depth of 190 mm. The core was excited at several inductions and the measurements were done at different locations by inserting some search coils. The comparison between the investigation results from this work and the simulation results obtained from the proposed method of this paper is presented in Fig. 5*a*.

The second measurements were performed by Zurek *et al.* [7] using a series of search coils enclosing every five strips. The tested specimens were toroidal cores wound from grain-oriented electrical steel (grade M4) with the inner radius of 35 mm, outer radius of 52 mm and height of 40 mm and were magnetised at 0.5 and 1.5 T.

The results from the proposed method are in good agreement with the measurements done in these papers and the comparisons implemented in Figs. 5*a* and *b*, clarify the accuracy of the proposed method.

4 Simulation results and discussion

In the previous section, the required methodology to enable the development of more accurate numerical models that describes well the distribution of flux density inside the magnetic core was presented in detail and validated with measurements presented in two other papers. A decrease of flux density from inner to outer side of the core is expected to be observed according to the difference in magnetic path length. The overall trend of maximum flux density is lower at outer segments rising towards the inner points. However, the distribution pattern is different at distinct magnetisation. At lower peak flux density, the core operates at the linear part of the magnetisation curve. In this region, μ_r has a large constant value. On the other hand, by increasing the exciting flux density, the inner parts of the core which have higher flux density enter the non-linear section of the magnetisation curve. Accordingly, on one hand lower magnetic length in the inner parts will push the flux to the inner parts. On the other hand, as can be seen in Fig. 2, the

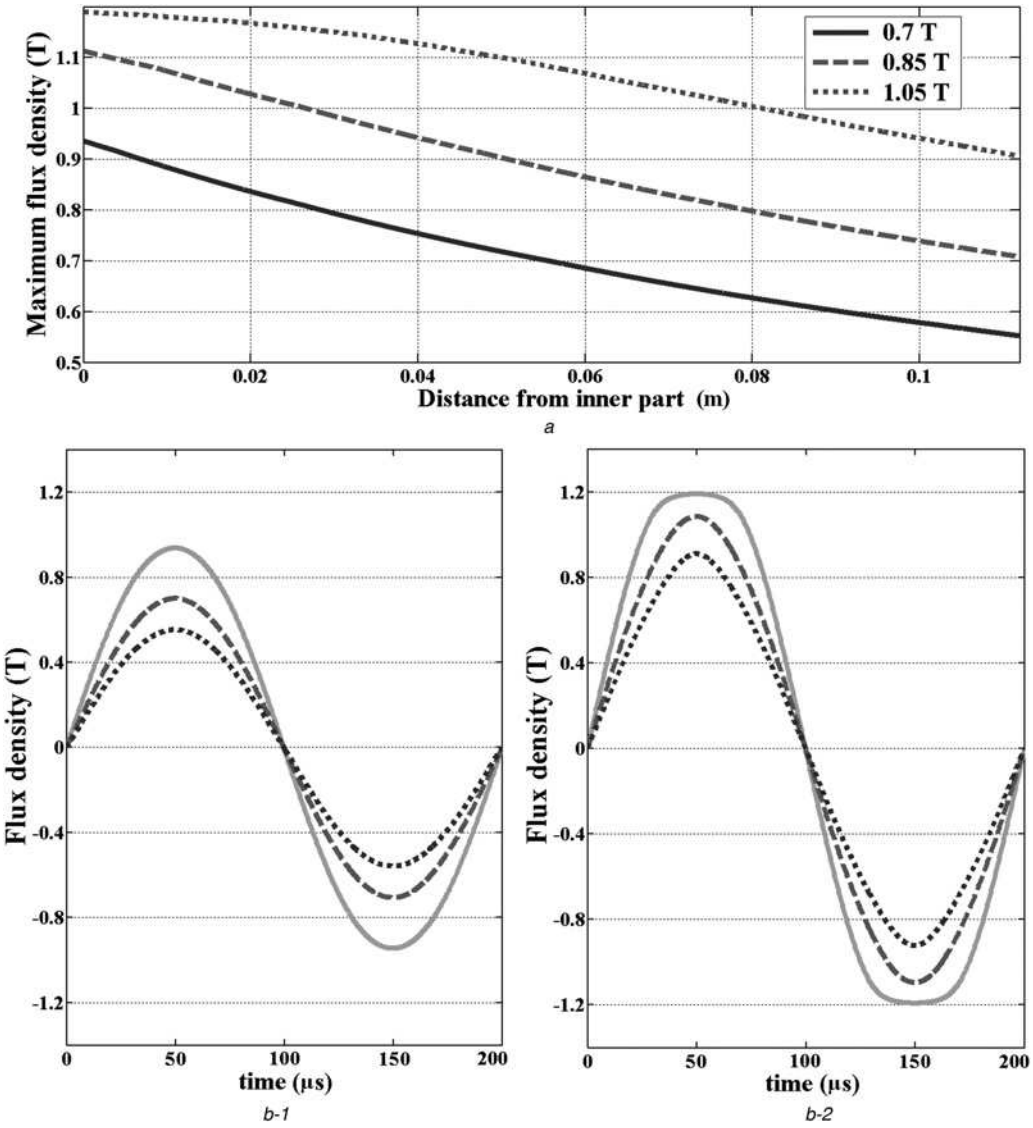


Fig. 6 *Pattern of flux density in the core*

a Distribution of B_{max} at three different excitations
b Waveform of flux density at two excitations (b-1 for 0.7T and b-2 for 1.05T), dotted line: outermost segment, dashed line: middle point, solid line: innermost segment

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permeability decreases at higher flux densities which increases the reluctance of the inner segments. These two phenomena have reverse impacts on the distribution of flux. Consequently, the variation pattern is different at higher inductions and the difference between the minimum and maximum values (ΔB) at low magnetisation is larger than the corresponding one at higher inductions. This effect is illustrated in Fig. 6*a* at several magnetisations. At 0.7 T, ΔB is approximately equal to 0.4 T, but at 1.05 T, it reduces to 0.3 T. Therefore this non-linear characteristic of the magnetic material have positive effect on the distribution of flux and make it more homogeneous. It can also be concluded from the waveform of flux density in the core. Figs. 6*b*-1 and 6*b*-2 illustrate the waveform of magnetic induction at different locations – inner, middle and outer – for two excitations, 0.7 and 1.05 T. As can be seen, all the waveforms for lower induction are almost pure sinusoidal, since they are in the linear section of *B*–*H* curve. On the contrary, at higher inductions, the inner segments approach saturation.

With respect to non-linear behaviour of the cores, magnetic loss varies according to the changes in the loss density ($B(x, t)$). The total core loss is the integration of $p(x, t)$ over the volume of the core.

$$p_t(t) = \int_0^a p(x, t) bl(x)\, \mathrm{d}x \tag{18}$$

where a and b are the geometrical dimensions of the core cross-section, $p_t(t)$ is instantaneous total loss of the core and $l(x)$ is the magnetic length at point x. Therefore the average total loss P_t will be

$$P_\mathrm{t} = \frac{1}{T}\int_0^\mathrm{T} p_\mathrm{t}(t)\ \mathrm{d}t \tag{19}$$

All the existing core loss calculation methods can be applied in determining $p(x, t)$.

To investigate the effect of flux density variation along the cross section of the core, a 800 kVA transformer was designed with nanocrystalline material based on the design procedures described in [20, 21]. The rated voltage and operating frequency are 1/10 kV and 5 kHz, respectively. The geometrical parameters and the number of primary turns are presented in Table 1.

The empirical Steinmetz equations were selected in this paper because of their simplicities and privileges for designers. These methods include OSE, MSE, GSE and IGSE which were expressed briefly in Section 2. In these approaches, the proposed method was used to take the flux changes into consideration. In addition, OSE was also implemented on the average constant value of peak flux density for the whole cross section of the core. The coefficients required for core estimation have been computed by several experiments implemented on thin ring cores made of VITROPERM 500 F material. These tests were done from 100 Hz to 10 kHz and for the flux density range of 0.1–1.1 T. The factors existing in Steinmetz

Table 1 Parameters of 800 kVA transformer

a, mm	*b*, mm	h_c, mm	*d*, mm	*N*
110	110	200	100	5

Table 2 I core loss calculation coefficients of VITROPERM 500 F

k	α	β
7.55e-7	1.98	2.27

methods were computed by curve fitting of test results using expression in (3) and are expressed in Table 2.

Using the obtained values and applying to the modelled single phase core type transformer, the localised magnetic loss density along the core $P_\mathrm{c}(x)$ was calculated. The relative results for two different excitations are shown in Figs. 7*a* and *b* (0.7 and 1.05 T, respectively). In these figures, OSE-1 relates to core loss density obtained by constant flux density and OSE-2 and other methods correspond to variation of flux density proposed in the paper. $P_\mathrm{c}(x)$ has reverse relation to the distance from inner section and will decrease from the inner to outer points. At 0.7 T, where B varies from 0.55 to 0.95 T according to Fig. 7*a*, most of the edited methods have similar accuracies. Nevertheless, at higher induction (1.05 T), which is above the knee point of B–*H* curve, because of non-linear behaviour of magnetic material and dependency of relative permeability on B, the relative difference decreases from 40% to lower than 30%.

At the innermost part, considering the variation of flux density results in approximately 1.5 times higher magnitude in comparison with the constant flux density (OSE-1). On the contrary, loss density at the outermost segment is 1.25 times lower than OSE-1. As can be seen in Figs. 7*a* and *b*, from the inner part to the middle of the core, P_c obtained by modified methods is higher than OSE-1 and for the rest of the core, P_c is lower. On this account, there would be a considerable difference in the generated loss of the inner half and outer half of the core. Consequently, a detailed study of core loss at several inductions was done on generated loss for inner, outer and total volume of the core and is presented in Table 3. The ratio of the losses of inner half to the outer one obtained from different methods against the magnetic induction is depicted in Fig. 8. Although the average loss density in the inner half is almost 2 times more than the outer one, but the corresponding volume is 30% lower. As the consequence, P_t in the inner half is 1.4 times higher. This ratio decreases by increasing the operating flux, since at higher B, the difference between the minimum and maximum of flux density will be lower. It can also be mentioned that the results acquired by OSE-1 are constant and equal to the ratio of volumes. The other fact which can be deduced from Fig. 8 is that at higher inductions where the inner sections enter the saturation, OSE-2 loses its accuracy, because of the distorted waveform of B.

The other fact which proves the importance of considering the variation of flux density is the maximum temperature inside the core and transformer. The temperature profile is strongly dependent on the spatial loss and heat generation. Thus, considering the inhomogeneous profile of core loss density would have significant effect on temperature rise inside the core.

Consider a solid heat conductor material as shown in Fig. 9*a* with heat conduction k_m and homogeneous volumetric heat generation $P_\mathrm{c}(x)$. The temperature and heat flux variation ($T(x)$ and $q(x)$) under unidirectional steady-state heat conduction are formulated for the control

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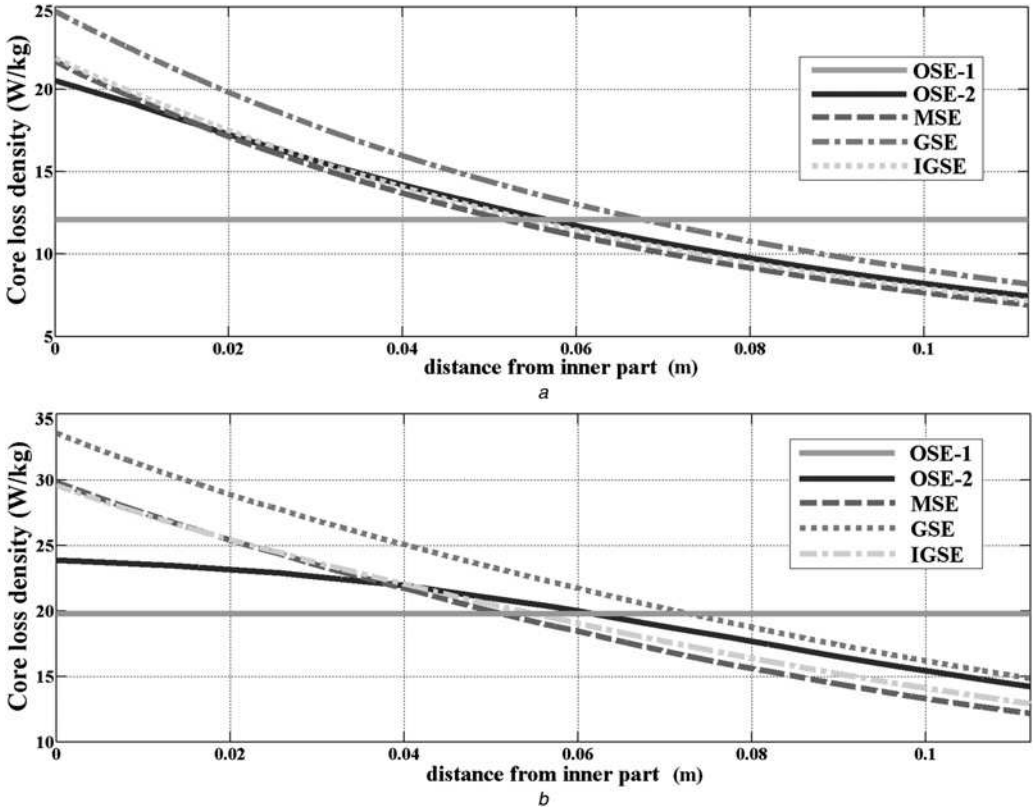


Fig. 7 *Calculated core loss density variation with empirical methods in the core at two different excitations*
a At 0.7T
b At 1.05T

Table 3 Loss division between the inner and outer half of the core at several excitations

B_m (T)	OSE-1			OSE-2			MSE			GSE			IGSE		
	Inner	Outer	Total	Inner	Outer	Total	Inner	Outer	Total	Inner	Outer	Total	Inner	Outer	Total
1.1	603.3	822.7	1426	711	737	1448	753	652	1405	862	784	1646	759	685	1444
1.05	542.8	740.2	1283	667	631	1298	687	569	1256	790	679	1469	696	594	1290
1	485.7	662.3	1148	616	543	1159	622	499	1121	717	592	1309	633	519	1152
0.9	382.5	521.5	904	503	407	910	497	383	880	576	450	1026	508	395	903
0.8	292.8	399.2	692	394	304	698	385	288	673	447	338	785	394	297	691
0.7	216.2	294.8	511	295	220	515	287	210	497	333	246	579	294	218	512
0.6	152.3	207.7	360	210	153	363	203	147	350	237	172	409	209	151	360

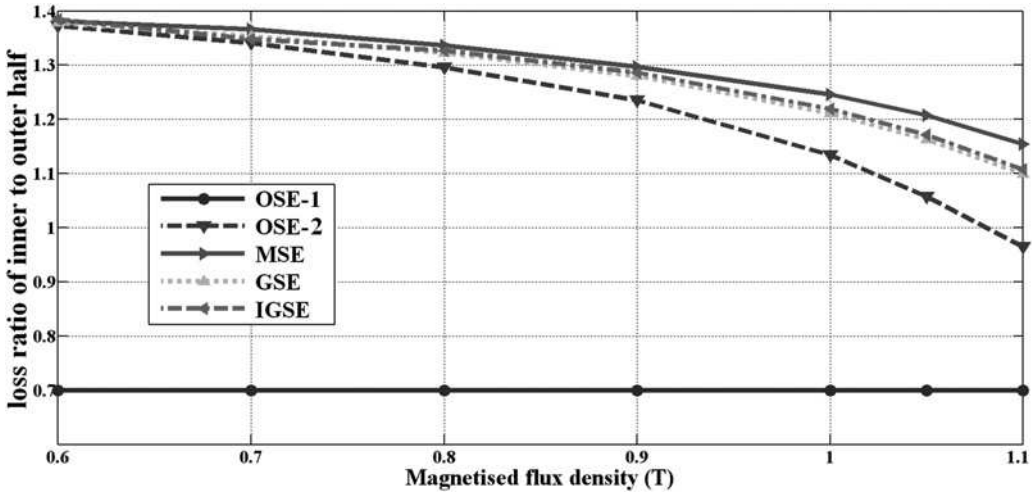


Fig. 8 *Ratio of the loss of inner half to outer one at several inductions*

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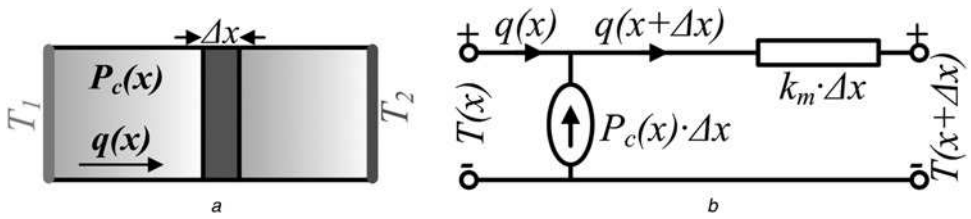


Fig. 9 *Heat conduction model*

a 1D heat conduction with volumetric loss
b Equivalent thermal-electric model for subsection Δx

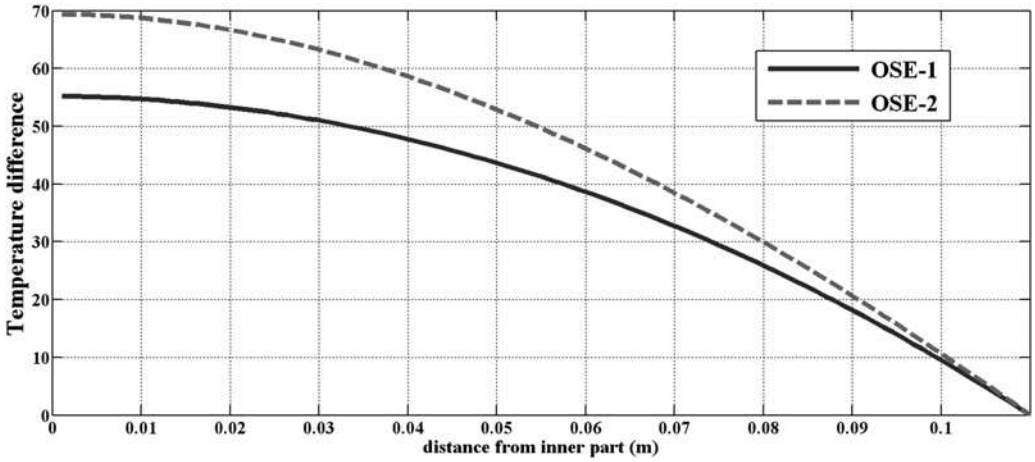


Fig. 10 *Comparison of temperature difference variation from the outer point obtained by conventional (OSE-1) and modified OSE (OSE-2)*

section Δx based on the equivalent thermal circuit illustrated in Fig. 9*b*.

$$\left.\begin{aligned}\frac{\mathrm{d}T(x)}{\mathrm{d}x} &= -k_{\mathrm{M}}q(x)\\ \frac{\mathrm{d}q(x)}{\mathrm{d}x} &= P_{\mathrm{c}}(x)\end{aligned}\right\} \Rightarrow \frac{\mathrm{d}^2T}{\mathrm{d}x^2} = -k_{\mathrm{M}}P_{\mathrm{c}}(x) \qquad (20)$$

According to (20), temperature variation depends directly on the localised magnitude of loss. In the previous existing thermal modelling expressions, $P_{\mathrm{c}}(x)$ was assumed to be constant. In Fig. 10, the profiles of temperature difference between the inner and outer temperature along the core obtained by conventional OSE (OSE-1) and modified OSE (OSE-2) using the proposed method are presented. Considering varying loss profile will result in 15°K higher for maximum temperature in the core.

5 Conclusion

The originality of this paper is to present an accurate, simple and fast method investigating the flux and core loss distribution inside single phase wound core of medium-frequency high-power transformers. The proposed method has several special features. First, it takes into account the non-linear characteristic of magnetic material. This non-linearity has significant effects on localised flux density. Second, the main typical core topologies of wound cores such as toroidal, shell and core types can be modelled. The simulation results produced by proposed approach showed a good correspondence with measurement results from the other papers.

Owing to non-linear behaviour and different magnetic lengths in the core, flux will be pushed to the inner segments. As a result, at high inductions, the inner segments go to saturation, while the outer segments are in linear part of B-H curve. Empirical Steinmetz equations were implemented to study the distribution of core loss. It has been demonstrated that at nominal flux density (e.g. 0.9 T for nanocrystalline), the inner half of the core generates 30% higher loss in spite of having 0.7 times lower volume comparing with the outer one. This inhomogeneous distribution of the magnetic loss results in higher maximum operating temperature in the core.

The authors believe that the proposed methodology may provide a precise approach to investigate the losses and thermal behaviour of all topologies of transformers subjected to arbitrary waveforms. In the subsequent paper, the method will be modified to be applied to three-phase transformers of three and five-leg topologies.

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